

Unit 13. Unsupervised Learning

1 Principal Components

ISLR §12.1–12.2.

There is no response. Every criterion the course has used, from the residual sum of squares of Unit 2 to the out-of-bag error of Unit 12, measured a prediction against something observed, and none of them is available. What can still be asked of $\mathbf{X}$ alone is whether its k columns carry k directions' worth of information, and whether its n rows fall into groups.

1.1 Two definitions, one answer

Unit 5 defined the loading vectors as maximising the variance of the scores and proved they are the eigenvectors of the sample covariance matrix $\mathbf{S} = \mathbf{X}'\mathbf{X}/(n - 1)$, with $\mathbf{X}$ centred. A second definition asks for the best low-dimensional approximation instead, and gives the same vectors.

Definition 13.1 (Rank- M approximation). For $1 \leq M \leq k$ let $\mathcal{A}_M$ be the set of $n \times k$ matrices of rank at most M . The best rank- M approximation problem is

$$\min_{\mathbf{A} \in \mathcal{A}_M} \|\mathbf{X} - \mathbf{A}\|_F^2, \quad \|\mathbf{B}\|_F^2 = \sum_{i,j} b_{ij}^2 = \text{tr}(\mathbf{B}'\mathbf{B}). \quad (1)$$

Theorem 13.2 (Eckart–Young). Let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}'$ be a singular value decomposition (prerequisites) with singular values $d_1 \geq \dots \geq d_k \geq 0$, and let $\mathbf{X}_M = \sum_{m \leq M} d_m \mathbf{u}_m \mathbf{v}_m'$ be the truncation. Then $\mathbf{X}_M$ solves (1) and

$$\min_{\mathbf{A} \in \mathcal{A}_M} \|\mathbf{X} - \mathbf{A}\|_F^2 = \sum_{m > M} d_m^2. \quad (2)$$

Its columns of $\mathbf{V}$ are the loading vectors of Unit 5 and $\mathbf{U}\mathbf{D}$ holds the scores, so the two definitions of a principal component coincide.

Proof. The value is attained. $\mathbf{X} - \mathbf{X}_M = \sum_{m > M} d_m \mathbf{u}_m \mathbf{v}_m'$, and the $\mathbf{u}_m$ are orthonormal, as are the $\mathbf{v}_m$, so $\text{tr}\{(\mathbf{X} - \mathbf{X}_M)'(\mathbf{X} - \mathbf{X}_M)\} = \sum_{m > M} d_m^2$.

No rank- M matrix does better. Let $\mathbf{A}$ have rank at most M and let $\mathbf{P}$ be the orthogonal projection on the column space of $\mathbf{A}$, of rank $r \leq M$. Since the columns of $\mathbf{A}$ lie in that space, $\mathbf{P}\mathbf{A} = \mathbf{A}$, and for each column $\mathbf{x}$ of $\mathbf{X}$ the projection theorem (prerequisites) gives $|\mathbf{x} - \mathbf{P}\mathbf{x}| \leq |\mathbf{x} - \mathbf{A}\mathbf{x}^*|$ for the corresponding column; summing over columns, $\|\mathbf{X} - \mathbf{A}\|_F^2 \geq \|\mathbf{X} - \mathbf{P}\mathbf{X}\|_F^2$. So it suffices to minimise over projections of rank M .

Write $\mathbf{P} = \mathbf{W}\mathbf{W}'$ with $\mathbf{W}$ having M orthonormal columns. Then

$$\|\mathbf{X} - \mathbf{P}\mathbf{X}\|_F^2 = \text{tr}(\mathbf{X}'\mathbf{X}) - \text{tr}(\mathbf{W}'\mathbf{X}'\mathbf{X}\mathbf{W}),$$

because $\text{tr}\{(\mathbf{I} - \mathbf{P})\mathbf{X}\mathbf{X}'(\mathbf{I} - \mathbf{P})\} = \text{tr}(\mathbf{X}'\mathbf{X}) - \text{tr}(\mathbf{P}\mathbf{X}\mathbf{X}')$ and $\mathbf{P}$ is idempotent. Maximising $\text{tr}(\mathbf{W}'\mathbf{X}'\mathbf{X}\mathbf{W})$ over orthonormal $\mathbf{W}$ is maximising $\sum_{m \leq M} \mathbf{w}_m' \mathbf{X}'\mathbf{X} \mathbf{w}_m$, which by the argument of Unit 5, applied one direction at a time, is at most $\sum_{m \leq M} d_m^2$, the d_m^2 being the eigenvalues of $\mathbf{X}'\mathbf{X}$; equality holds at $\mathbf{W} = [\mathbf{v}_1 \cdots \mathbf{v}_M]$. Subtracting from $\text{tr}(\mathbf{X}'\mathbf{X}) = \sum_m d_m^2$ gives (2).

The two definitions agree. $\mathbf{X}'\mathbf{X} = \mathbf{V}\mathbf{D}^2\mathbf{V}'$, so the eigenvectors of $\mathbf{S} = \mathbf{X}'\mathbf{X}/(n-1)$ are the columns of $\mathbf{V}$ with eigenvalues $\lambda_m = d_m^2/(n-1)$, which is the statement of Unit 5, and the scores are $\mathbf{X}\mathbf{v}_m = d_m\mathbf{u}_m$. $\square$

Remark. So the first M components are simultaneously the M directions of greatest variance and the M -dimensional plane closest to the data in squared distance. That the two questions have one answer is what makes the components worth looking at: a picture of the first two scores is both the most spread-out projection and the least distorted one.

Corollary 13.3 (Variance explained is error removed). With $\lambda_m = d_m^2/(n-1)$,

$$\frac{\sum_{m \leq M} \lambda_m}{\sum_{m \leq k} \lambda_m} = 1 - \frac{\|\mathbf{X} - \mathbf{X}_M\|_F^2}{\|\mathbf{X}\|_F^2}, \quad (3)$$

so the cumulative proportion of variance explained of Unit 5 is one minus the relative approximation error.

Proof. Divide (2) by $\|\mathbf{X}\|_F^2 = \text{tr}(\mathbf{X}'\mathbf{X}) = \sum_m d_m^2$ and use $\lambda_m \propto d_m^2$. □

1.2 What is not determined

Proposition 13.4 (Sign and rotation). (i) If $\mathbf{v}_m$ is an m th loading vector so is $-\mathbf{v}_m$, with scores negated; every statement about variance, approximation and PVE is unchanged.

(ii) If $\lambda_m = \lambda_{m+1}$ then any orthonormal pair spanning the same two-dimensional eigenspace serves, so the individual components are not determined at all.

Proof. (i) $(-\mathbf{v})'\mathbf{S}(-\mathbf{v}) = \mathbf{v}'\mathbf{S}\mathbf{v}$ and $\mathbf{X}(-\mathbf{v}) = -\mathbf{X}\mathbf{v}$; the rank- M approximation $\sum_m d_m \mathbf{u}_m \mathbf{v}_m'$ is unchanged when both $\mathbf{u}_m$ and $\mathbf{v}_m$ change sign.

(ii) If $\mathbf{S}\mathbf{w} = \lambda\mathbf{w}$ for every $\mathbf{w}$ in a two-dimensional space, any orthonormal basis of it consists of eigenvectors with the same eigenvalue, and the argument of Unit 5 selects none of them over another. $\square$

Remark. Reading a loading vector as though its signs meant something is therefore reading an arbitrary choice made by the software. What is determined is the subspace and the ordering of the eigenvalues, and any interpretation must be invariant under the sign flip: that Assault and Murder load together is a statement, that they load positively is not.

Remark. Unit 5 recorded that $\mathbf{S}$ changes under rescaling of the columns and that standardising is the usual default. Nothing above weakens that: Theorem 13.2 minimises squared error in whatever units the columns carry, so the choice of units is the choice of what the approximation is asked to preserve. §5.1 shows the first component changing completely.

1.3 How many components

Definition 13.5 (Scree plot). The plot of $\text{PVE}_m = \lambda_m / \sum_l \lambda_l$ against m , or of its cumulative sum.

Remark. There is no criterion here of the kind Unit 5 had. Cross-validation compares predictions and there is nothing to predict; the sequence $\sum_{m \leq M} \lambda_m$ increases in M by construction, so it cannot be optimised over M any more than the training error of Unit 11 could be optimised over tree size. What is done instead is to look for an elbow, which is a judgement, and to check that the components kept admit a reading. When the components feed a supervised model, as in the principal components regression of Unit 5, M becomes a tuning parameter again and cross-validation returns.

Definition 13.6 (Biplot). The scatter of the first two score columns with the first two rows of $\mathbf{V}'$ drawn as arrows from the origin on a common pair of axes.

Proposition 13.7 (What the arrows mean). The j th arrow is (v_{j1}, v_{j2}) , and the projection of a score point (z_{i1}, z_{i2}) on that arrow's direction is the part of x_{ij} reconstructed by the first two components: $\hat{x}_{ij} = z_{i1}v_{j1} + z_{i2}v_{j2}$. Two arrows at a small angle belong to variables whose reconstructions are nearly proportional across observations.

Proof. The rank-two reconstruction is $\mathbf{X}_2 = \sum_{m \leq 2} d_m \mathbf{u}_m \mathbf{v}_m'$, whose (i, j) entry is $\sum_{m \leq 2} z_{im} v_{jm}$ with $z_{im} = d_m u_{im}$. Reading that as an inner product of (z_{i1}, z_{i2}) with (v_{j1}, v_{j2}) gives the first claim, and the cosine of the angle between two arrows is the correlation of the two reconstructions when the score columns are scaled to equal norm. $\square$

2 K-means Clustering

ISLR §12.4.1.

Definition 13.8 (The criterion). A **clustering** into K groups is a partition $C_1, \dots, C_K$ of

$\{1, \dots, n\}$ with every C_g non-empty. Its **within-cluster sum of squares** is

$$W(C) = \sum_{g=1}^K \sum_{i \in C_g} |\mathbf{x}_i - \bar{\mathbf{x}}_g|^2, \quad \bar{\mathbf{x}}_g = \frac{1}{|C_g|} \sum_{i \in C_g} \mathbf{x}_i, \quad (4)$$

and K -means clustering is the problem $\min_C W(C)$.

- Proposition 13.9** (Three readings of the criterion). (i) $\sum_{i \in C_g} |\mathbf{x}_i - \bar{\mathbf{x}}_g|^2 = \frac{1}{2|C_g|} \sum_{i \in C_g} \sum_{i' \in C_g} |\mathbf{x}_i - \mathbf{x}_{i'}|^2$: the within-cluster sum of squares is a sum of pairwise distances and never refers to a centre.
- (ii) $W(C) + B(C) = T$, where $T = \sum_i |\mathbf{x}_i - \bar{\mathbf{x}}|^2$ is free of C and $B(C) = \sum_g |C_g| |\bar{\mathbf{x}}_g - \bar{\mathbf{x}}|^2$. Minimising W is maximising B .
- (iii) For fixed centres $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$, $\sum_g \sum_{i \in C_g} |\mathbf{x}_i - \boldsymbol{\mu}_g|^2$ is minimised over partitions by assigning each i to a nearest centre; and for a fixed partition it is minimised over centres at $\boldsymbol{\mu}_g = \bar{\mathbf{x}}_g$.

Proof. (i) Expanding, $\sum_{i,i' \in C_g} |\mathbf{x}_i - \mathbf{x}_{i'}|^2 = \sum_{i,i'} \{|\mathbf{x}_i|^2 - 2\mathbf{x}_i' \mathbf{x}_{i'} + |\mathbf{x}_{i'}|^2\} = 2|C_g| \sum_i |\mathbf{x}_i|^2 - 2|\sum_i \mathbf{x}_i|^2 = 2|C_g| \{ \sum_i |\mathbf{x}_i|^2 - |C_g| |\bar{\mathbf{x}}_g|^2 \}$, and the brace is $\sum_{i \in C_g} |\mathbf{x}_i - \bar{\mathbf{x}}_g|^2$.

(ii) is the analysis of variance identity of Unit 10 applied coordinatewise, the cross terms vanishing because $\sum_{i \in C_g} (\mathbf{x}_i - \bar{\mathbf{x}}_g) = \mathbf{0}$.

(iii) The double sum splits over i , and the term for i is $|\mathbf{x}_i - \boldsymbol{\mu}_{g(i)}|^2$, smallest when $g(i)$ indexes a nearest centre. For fixed C the sum splits over g and each part is minimised at the mean (Unit 2). $\square$

Definition 13.10 (Lloyd's algorithm). Start from an assignment. Repeat: compute the cluster means; reassign each observation to a nearest mean. Stop when the assignment does not change.

Theorem 13.11 (The algorithm decreases the criterion and stops). Write $C^{(t)}$ for the assignment after t rounds. Then $W(C^{(t+1)}) \leq W(C^{(t)})$, and the algorithm reaches an assignment it does not change after finitely many rounds.

Proof. Let $\boldsymbol{\mu}_g^{(t)}$ be the means of $C^{(t)}$ and put $F(C, \boldsymbol{\mu}) = \sum_g \sum_{i \in C_g} |\mathbf{x}_i - \boldsymbol{\mu}_g|^2$. Then

$$W(C^{(t)}) = F(C^{(t)}, \boldsymbol{\mu}^{(t)}) \geq F(C^{(t+1)}, \boldsymbol{\mu}^{(t)}) \geq F(C^{(t+1)}, \boldsymbol{\mu}^{(t+1)}) = W(C^{(t+1)}),$$

the first inequality by the assignment half of Proposition 13.9(iii) and the second by the centre half, and the two equalities because $W(C) = \min_{\boldsymbol{\mu}} F(C, \boldsymbol{\mu})$ attained at the cluster means.

There are finitely many partitions of n points into K groups, and W never increases, so the sequence $W(C^{(t)})$ is non-increasing and takes finitely many values, hence is eventually constant. Once $W(C^{(t+1)}) = W(C^{(t)})$ both inequalities are equalities, so $C^{(t+1)}$ is also optimal for $\boldsymbol{\mu}^{(t)}$ and the assignment step returns it again. $\square$

Remark. The theorem promises a fixed point of the algorithm and nothing more. It is not a global minimum: the number of partitions is a Stirling number of the second kind, which for $n = 50$ and $K = 3$ exceeds 10^{23} , and no algorithm here searches them. The remedy is to run Lloyd's algorithm from many random starts and keep the best, and §4.3 shows the spread of values reached.

Proposition 13.12 (W cannot choose K). The optimal W is non-increasing in K , and equals 0 at $K = n$.

Proof. Any partition into K groups is refined into one with $K + 1$ by splitting a group with at least two members, and by Proposition 13.9(ii) applied within that group the within-cluster sum of squares cannot rise. At $K = n$ every cluster is a single point and each term of (4) vanishes. $\square$

3 Hierarchical Clustering

ISLR §12.4.2.

K -means needs K before it starts and returns one partition. Agglomerative clustering returns a nested family of partitions, one for each K , and asks only for a dissimilarity.

Definition 13.13 (Agglomerative clustering and linkage). Begin with n singleton clusters and a dissimilarity $d(i, i')$. At each step merge the two clusters with the smallest **linkage** D , and record

the merge height D . The usual linkages, for clusters A and B , are

$$\text{single: } D(A, B) = \min_{i \in A, i' \in B} d(i, i'), \quad (5)$$

$$\text{complete: } D(A, B) = \max_{i \in A, i' \in B} d(i, i'), \quad (6)$$

$$\text{average: } D(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{i' \in B} d(i, i'), \quad (7)$$

$$\text{centroid: } D(A, B) = |\bar{\mathbf{x}}_A - \bar{\mathbf{x}}_B|. \quad (8)$$

The **dendrogram** draws each merge as a join at its height.

Proposition 13.14 (Cutting the dendrogram). The clusters present at any moment form a partition, and every cluster at a later moment is a union of clusters at an earlier one. Cutting the dendrogram at height h therefore returns a partition, and the partitions obtained at different heights are nested.

Proof. The initial clusters partition the observations, and replacing two blocks by their union leaves a partition; each step does that once, which is also the nesting claim. $\square$

Theorem 13.15 (No inversions). Suppose the linkage satisfies the **reducibility** inequality

$$D(A \cup B, C) \geq \min\{D(A, C), D(B, C)\} \quad (9)$$

for all disjoint clusters. Then the merge heights are non-decreasing, so the dendrogram has no inversion. Single, complete and average linkage all satisfy (9).

Proof. Let h be the height of the current merge, of A with B , so $D(A, B) = h$ and every other pair of current clusters has linkage at least h . After the merge the available pairs are of two kinds. A pair not involving $A \cup B$ has the same linkage as before, hence at least h . A pair $(A \cup B, C)$ has, by (9), linkage at least $\min\{D(A, C), D(B, C)\} \geq h$, both terms having been available before the merge. So the next minimum is at least h .

Single linkage attains equality in (9), the minimum over $A \cup B$ being the smaller of the two minima. Complete linkage gives $\max\{D(A, C), D(B, C)\} \geq \min\{D(A, C), D(B, C)\}$. Average

linkage gives $\{|A|D(A, C) + |B|D(B, C)\}/(|A| + |B|)$, a weighted mean of the two, hence at least their minimum. $\square$

Remark. Centroid linkage is not of this form: $\bar{\mathbf{x}}_{A \cup B}$ is a weighted mean of $\bar{\mathbf{x}}_A$ and $\bar{\mathbf{x}}_B$ and can be strictly closer to $\bar{\mathbf{x}}_C$ than either, so (9) fails and a merge can occur below the height of the merge that produced it. The dendrogram then crosses itself and cutting it at a height no longer returns the clusters the picture suggests. §4.4 counts seven such inversions on the course data set, and none for the other three linkages.

Proposition 13.16 (Single linkage chains). Under single linkage, two observations are in the same cluster at height h if and only if they are joined by a chain $i = i_0, i_1, \dots, i_r = i'$ with $d(i_{l-1}, i_l) \leq h$ at every step.

Proof. Merging at height at most h requires only one pair at distance at most h , so the clusters at height h are exactly the connected components of the graph on $\{1, \dots, n\}$ whose edges are the pairs at distance at most h , and membership of a component is the existence of such a chain. $\square$

Remark. That is single linkage's characteristic behaviour and its characteristic failure. A sequence of intermediate points at small spacing joins two otherwise distant groups, so single linkage finds elongated clusters that the other linkages break, and one stray point between two clusters merges them. Complete linkage, which is governed by the farthest pair, produces compact clusters and splits elongated ones.

Remark. Nothing here says which linkage or which dissimilarity is right, and there is no criterion to appeal to: the choices define what a cluster is. Euclidean distance on standardised columns and correlation-based dissimilarity answer different questions, the first about magnitudes and the second about profiles, and two observations can be close in one and far in the other.

4 Computation: Formulas Against Package Output

ISLR §12.5.1, §12.5.3.

USArrests from **ISLR2**: four crime and population measurements on the fifty states, in four different units, which is the case standardisation exists for.

```
1 suppressMessages(library(ISLR2))  
2 X <- scale(USArrests); n <- nrow(X); k <- ncol(X)
```

```
3 round(rbind(mean = colMeans(USArrests), sd = apply(USArrests, 2, sd)), 2)
```

```
##           Murder Assault UrbanPop Rape
## mean      7.79   170.76     65.54 21.23
## sd        4.36    83.34     14.47 9.37
```

4.1 Principal components (§1.1)

Three routes to the same components: the package, the eigendecomposition of the correlation matrix, and the singular value decomposition of the standardised data.

```
1 pr <- prcomp(USArrests, scale. = TRUE)
2 E <- eigen(cor(USArrests)); sv <- svd(X)
3 round(rbind(prcomp_sdev = pr$sdev, sqrt_eigenvalue = sqrt(E$values),
4             singular_over_sqrt_n_minus_1 = sv$d/sqrt(n - 1)), 6)
```

```
##                [,1]    [,2]    [,3]    [,4]
## prcomp_sdev      1.575 0.9949 0.5971 0.4164
## sqrt_eigenvalue  1.575 0.9949 0.5971 0.4164
## singular_over_sqrt_n_minus_1 1.575 0.9949 0.5971 0.4164
```

```
1 c(loadings_agree_to = max(abs(abs(pr$rotation) - abs(E$vectors))),
2   scores_agree_to = max(abs(abs(pr$x) - abs(X %*% E$vectors))),
```

```
3 pve_sums_to = sum(pr$sdev^2)/k, trace_of_correlation = sum(diag(cor(USArrests))))
```

```
##      loadings_agree_to      scores_agree_to
##      1.554e-15            3.164e-15
##      pve_sums_to trace_of_correlation
##      1.000e+00            4.000e+00
```

The comparison is of absolute values, because by Proposition 13.4(i) the sign of each column is arbitrary. On this data the two routines happen to agree; nothing makes them, and §5.2 exhibits the same call returning the opposite sign after the rows are permuted.

```
1 rbind(prcomp = round(pr$rotation[, 1], 4), eigen = round(E$vectors[, 1], 4))
```

```
##      Murder Assault UrbanPop      Rape
## prcomp -0.5359 -0.5832  -0.2782 -0.5434
## eigen  -0.5359 -0.5832  -0.2782 -0.5434
```

Theorem 13.2: the squared approximation error of the rank- M truncation equals the sum of the discarded squared singular values, and Corollary 13.3 turns that into the cumulative proportion of variance.

```
1 recon <- function(M) if (M == 0) matrix(0, n, k) else
```

```

2   sv$u[, 1:M, drop = FALSE] %*% diag(sv$d[1:M], M, M) %*% t(sv$v[, 1:M, drop = FALSE])
3   t(sapply(0:4, function(M) c(M = M,
4     squared_error = sum((X - recon(M))^2),
5     dropped_singular_squares = sum(sv$d[seq_len(k) > M]^2),
6     one_minus_relative_error = 1 - sum((X - recon(M))^2)/sum(X^2),
7     cumulative_pve = sum(pr$sdev[seq_len(M)]^2)/k)))

```

```

##           M squared_error dropped_singular_squares
## [1,] 0      1.960e+02      196.000
## [2,] 1      7.447e+01      74.468
## [3,] 2      2.597e+01      25.970
## [4,] 3      8.498e+00      8.498
## [5,] 4      9.337e-29      0.000
##           one_minus_relative_error cumulative_pve
## [1,]           0.0000           0.0000
## [2,]           0.6201           0.6201
## [3,]           0.8675           0.8675
## [4,]           0.9566           0.9566
## [5,]           1.0000           1.0000

```

That the truncation is optimal, and not merely one rank- M matrix among many, is checked by

minimising (1) directly: alternating least squares over the two factors from twenty random starts.

```
1 als <- function(M, iter = 200) { V <- matrix(rnorm(k*M), k, M)
2   for (it in 1:iter) { U <- X %*% V %*% solve(crossprod(V))
3     V <- t(X) %*% U %*% solve(crossprod(U)) }
4   sum((X - U %*% t(V))^2) }
5 set.seed(2); v <- replicate(20, als(2))
6 c(best_of_twenty_starts = min(v), worst = max(v),
7   truncated_svd = sum((X - recon(2))^2))
```

```
## best_of_twenty_starts          worst
##                25.97          25.97
##      truncated_svd
##                25.97
```

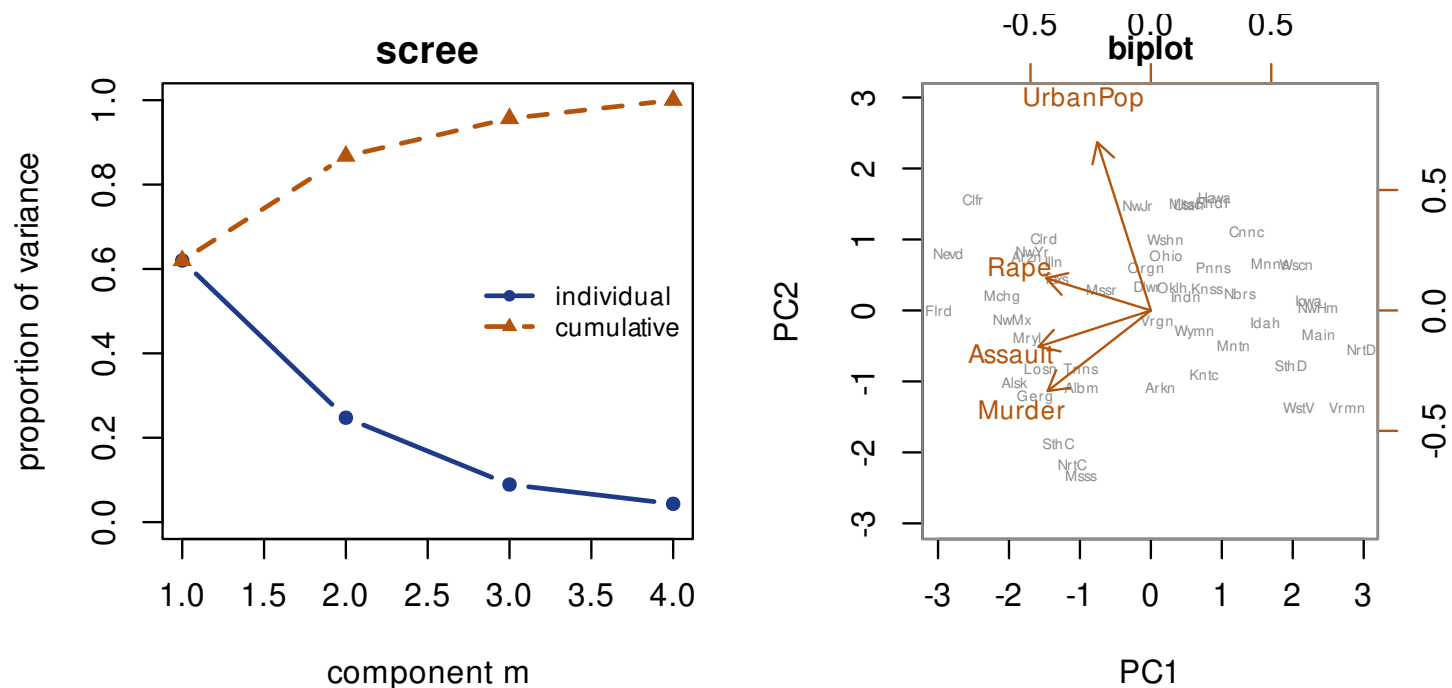


Figure 13.1: Left: the scree plot of Definition 13.5; two components carry 87% of the variance, which by Corollary 13.3 is the same statement as a relative approximation error of 13%. Right: the biplot of Definition 13.6. Three crime arrows point nearly together and **UrbanPop** nearly at right angles to them, so the second component separates urbanisation from crime.

4.2 Scaling changes the answer (§1.2)

```

1 pu <- prcomp(USArrests, scale. = FALSE)
2 round(rbind(unscaled_loading_1 = pu$rotation[, 1],
3             scaled_loading_1 = pr$rotation[, 1],
4             column_sd = apply(USArrests, 2, sd)), 4)

```

```

##                Murder Assault UrbanPop    Rape
## unscaled_loading_1  0.0417  0.9952   0.0463  0.0752
## scaled_loading_1   -0.5359 -0.5832  -0.2782 -0.5434
## column_sd          4.3555 83.3377  14.4748  9.3664

```

```

1 c(unscaled_pve_1 = pu$sdev[1]^2/sum(pu$sdev^2),
2   scaled_pve_1 = pr$sdev[1]^2/sum(pr$sdev^2))

```

```

## unscaled_pve_1    scaled_pve_1
##           0.9655           0.6201

```

Unstandardised, the first component is **Assault** almost alone, because **Assault** is recorded in counts with a standard deviation of 83 against **Murder**'s 4.4, and it carries 97% of the variance. Standardised, all four variables contribute and the first component carries 62%.

4.3 *K*-means (§2)

Proposition 13.9, all three parts, and Theorem 13.11.

```

1 set.seed(4); km <- kmeans(X, centers = 3, nstart = 50)
2 wss <- function(lab) sum(sapply(unique(lab), function(g) {
3   Z <- X[lab == g, , drop = FALSE]; sum(scale(Z, scale = FALSE)^2) })))
4 g1 <- X[km$cluster == 1, ]
5 c(package_tot_withinss = km$tot.withinss, manual = wss(km$cluster),
6   total = sum(scale(X, scale = FALSE)^2),
7   within_plus_between = km$tot.withinss + km$betweenss,
8   pairwise_form = sum(as.matrix(dist(g1))^2)/nrow(g1),
9   twice_within_cluster_1 = 2*sum(scale(g1, scale = FALSE)^2))

```

```

##      package_tot_withinss              manual
##                78.32                78.32
##                total      within_plus_between
##                196.00                196.00
##                pairwise_form twice_within_cluster_1
##                39.25                39.25

```

Lloyd's algorithm coded from Definition 13.10, printing the criterion at each round.

```

1 lloyd <- function(Xm, K, seed) { set.seed(seed)
2   lab <- sample(K, nrow(Xm), replace = TRUE); tr <- numeric(0)
3   for (it in 1:30) {
4     C <- t(sapply(1:K, function(g) colMeans(Xm[lab == g, , drop = FALSE])))
5     D <- sapply(1:K, function(g) rowSums(sweep(Xm, 2, C[g, ]^2))
6     lab <- max.col(-D); tr <- c(tr, sum(apply(D, 1, min))) }
7   tr }
8 round(lloyd(X, 3, 11)[1:8], 3)

```

```
## [1] 174.91 136.14 93.63 85.44 82.05 81.59 81.59 81.59
```

The sequence is non-increasing and constant from the sixth round, as Theorem 13.11 requires. It is a fixed point and not a minimum: repeated single starts land in several places.

```

1 set.seed(6)
2 one <- replicate(200, kmeans(X, centers = 3, nstart = 1)$tot.withinss)
3 c(best = min(one), worst = max(one), distinct_values = length(unique(round(one, 6))),
4   proportion_at_best = mean(round(one, 6) == round(min(one), 6)),
5   with_nstart_50 = kmeans(X, centers = 3, nstart = 50)$tot.withinss)

```

```
##           best           worst   distinct_values
```

```
##              78.32              86.96              4.00
## proportion_at_best      with_nstart_50
##              0.52              78.32
```

Proposition 13.12: the criterion falls with K whatever the data, so it cannot choose K .

```
1 set.seed(7)
2 t(sapply(1:8, function(K) c(K = K,
3   within = kmeans(X, centers = K, nstart = 25)$tot.withinss)))
```

```
##      K within
## [1,] 1 196.00
## [2,] 2 102.86
## [3,] 3  78.32
## [4,] 4  56.40
## [5,] 5  48.94
## [6,] 6  42.83
## [7,] 7  38.30
## [8,] 8  34.11
```

4.4 Hierarchical clustering (§3)

Theorem 13.15: the three reducible linkages have non-decreasing merge heights and centroid linkage does not.

```

1 dd <- dist(X)
2 t(sapply(c("single", "complete", "average", "centroid"), function(m) {
3   h <- hclust(dd, method = m)
4   c(smallest_height_step = min(diff(h$height)),
5     inversions = sum(diff(h$height) < 0),
6     largest_height = max(h$height)) })))

```

##	smallest_height_step	inversions	largest_height
## single	0.00154996	0	2.058
## complete	0.00447978	0	6.077
## average	0.00008469	0	3.322
## centroid	-0.13000704	7	2.322

Proposition 13.14: cutting at a height gives a partition, and the partitions nest.

```

1 hc <- hclust(dd, method = "complete")
2 tab <- table(cut2 = cutree(hc, 2), cut4 = cutree(hc, 4))

```

```
3 tab
```

```
##      cut4
## cut2  1  2  3  4
##      1  8 11  0  0
##      2  0  0 21 10
```

```
1 c(each_four_cluster_lies_in_one_two_cluster = all(colSums(tab > 0) == 1))
```

```
## each_four_cluster_lies_in_one_two_cluster
##                                     TRUE
```

Proposition 13.16: under single linkage the clusters at height h are the connected components of the graph of pairs closer than h .

```
1 hs <- hclust(dd, method = "single"); h0 <- hs$height[n - 4]
2 lab <- cutree(hs, h = h0 + 1e-9)
3 A <- as.matrix(dd) <= h0 + 1e-9
4 comp <- rep(0, n); cc <- 0
5 for (i in 1:n) if (comp[i] == 0) { cc <- cc + 1; front <- i
6   while (length(front)) { comp[front] <- cc
7     front <- setdiff(which(apply(A[front, , drop = FALSE], 2, any)),
```

```
8           which(comp > 0)) } }  
9 c(height = h0, clusters_from_cutree = length(unique(lab)),  
10    connected_components = cc,  
11    same_partition = all(table(lab, comp) %*% rep(1, cc) ==  
12                      apply(table(lab, comp), 1, max)))
```

```
##           height clusters_from_cutree  
##           1.241           4.000  
## connected_components      same_partition  
##           4.000           1.000
```



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4.5 The two methods on the same data

```
1 set.seed(9)
2 km3 <- kmeans(X, centers = 3, nstart = 50)$cluster
3 hc3 <- cutree(hclust(dd, "complete"), 3)
4 table(kmeans = km3, complete_linkage = hc3)
```

```
##           complete_linkage
## kmeans  1  2  3
##      1  8 11  1
##      2  0  0 17
##      3  0  0 13
```

```
1 c(within_kmeans = wss(km3), within_hclust = wss(hc3))
```

```
## within_kmeans within_hclust
##           78.32           84.40
```

K-means minimises (4) directly and hierarchical clustering does not, so the first reaches the smaller value; the two partitions nonetheless agree on most states.

5 Pitfalls

5.1 Not standardising, or standardising without asking

§4.2 showed the first component of **USArrests** changing from **Assault** alone to a balanced combination. The same choice governs clustering, because every dissimilarity in Definition 13.13 is computed in whatever units the columns carry.

```
1 cl_raw <- cutree(hclust(dist(USArrests), "complete"), 3)
2 cl_std <- cutree(hclust(dist(X), "complete"), 3)
3 table(unstandardised = cl_raw, standardised = cl_std)
```

```
##                standardised
## unstandardised  1  2  3
##                1  6  9  1
##                2  2  2 10
##                3  0  0 20
```

```
1 c(states_in_the_diagonal_cells = sum(diag(table(cl_raw, cl_std))),
2   n = nrow(USArrests))
```

```
## states_in_the_diagonal_cells          n
##                          28          50
```

The cluster labels are arbitrary, so the table is the object to read and not the diagonal: whichever

way the three groups of one analysis are matched to the three of the other, a large minority of the fifty states changes group. The two analyses are not the same analysis.

Diagnostic. A clustering or a component dominated by whichever variable happens to be recorded on the largest scale. **Fix.** Standardise unless the columns are already commensurable, and say which was done. Neither choice is right in general: standardising declares all four variables equally important, and not standardising declares that a count of assaults matters more than a murder rate because it is a bigger number.

5.2 Reading the signs of a loading vector

```
1 p1 <- prcomp(USArrests, scale. = TRUE)
2 p2 <- prcomp(USArrests[sample(nrow(USArrests)), ], scale. = TRUE)
3 rbind(first_run = round(p1$rotation[, 1], 4),
4        after_reordering_rows = round(p2$rotation[, 1], 4))
```

	Murder	Assault	UrbanPop	Rape
## first_run	-0.5359	-0.5832	-0.2782	-0.5434
## after_reordering_rows	0.5359	0.5832	0.2782	0.5434

Diagnostic. A sentence that would be reversed by multiplying a loading vector by -1 . **Fix.** By

Proposition 13.4(i) the sign is a convention of the routine and carries no information; report relations among the loadings, which the sign flip preserves, and not their direction. The same applies to the axes of a biplot.

5.3 Choosing K by the within-cluster criterion

Diagnostic. A K chosen as the minimiser of the within-cluster sum of squares, which by Proposition 13.12 is always $K = n$; §4.3 prints the sequence falling monotonically to $K = 8$ with no sign of stopping. **Fix.** There is no criterion of the supervised kind here, because there is nothing held out to predict. What is available is an elbow, which is a judgement, and stability under resampling. When the clustering feeds something with a response, K becomes a tuning parameter and cross-validation applies to that.

5.4 Reporting one run of K -means

```
1 set.seed(12)
2 r <- t(replicate(60, { z <- kmeans(X, centers = 4, nstart = 1)
3   c(z$tot.withinss, length(unique(z$cluster))) })))
4 c(best = min(r[, 1]), worst = max(r[, 1]),
5   spread_percent = 100*(max(r[, 1]) - min(r[, 1]))/min(r[, 1]),
```

```
6 distinct_optima = length(unique(round(r[, 1], 6))))
```

```
##                best                worst  spread_percent
##                56.40                75.92                34.60
## distinct_optima
##                5.00
```

Diagnostic. A K -means result reported without `nstart`, or two analysts getting different clusters from the same data. **Fix.** Theorem 13.11 guarantees a fixed point and nothing more, and the starts here reach several distinct values spanning a wide range. Use many starts and report the number used.

5.5 Cutting a dendrogram that has inversions

```
1 hcen <- hclust(dd, "centroid")
2 bad <- which(diff(hcen$height) < 0)
3 rbind(merge_index = bad[1:3],
4       height_before = round(hcen$height[bad[1:3]], 4),
5       height_after = round(hcen$height[bad[1:3] + 1], 4))
```

```
##                [,1]    [,2]    [,3]
```

```
## merge_index    3.0000 11.0000 22.0000
## height_before  0.4288  0.6195  0.7922
## height_after   0.4179  0.5660  0.6962
```

```
1 c(total_inversions = sum(diff(hcen$height) < 0))
```

```
## total_inversions
##                7
```

```
1 list(cut_by_height = tryCatch(length(unique(cutree(hcen,
2     h = hcen$height[bad[1]]))), error = function(e) conditionMessage(e)),
3     cut_by_number = length(unique(cutree(hcen, k = 3))))
```

```
## $cut_by_height
## [1] "the 'height' component of 'tree' is not sorted (increasingly)"
##
## $cut_by_number
## [1] 3
```

R declines to cut by height at all, which is the honest response: with the heights not sorted there is no partition for a horizontal line to name. Cutting by number of clusters still works, because that reads the merge order rather than the heights.

Diagnostic. A dendrogram whose branches double back, and a horizontal cut that either fails or returns a different number of clusters from the one the picture suggests. **Fix.** Centroid linkage fails the reducibility inequality (9) of Theorem 13.15, so its heights are not monotone and the usual reading of a cut is not available. Use single, complete or average linkage, all three of which the theorem covers, or cut by number of clusters rather than by height.

6 Exercises

Exercise 13.1. A standardised $n \times 5$ data matrix has singular values 3.2, 2.1, 1.4, 0.9, 0.4. What proportion of variance do the first two components explain, and what is the relative squared error of the best rank-two approximation?

Solution. The squared singular values are 10.24, 4.41, 1.96, 0.81, 0.16, summing to 17.58. The first two explain $(10.24 + 4.41)/17.58 = 0.833$.

By Theorem 13.2 the squared error of the best rank-two approximation is $1.96 + 0.81 + 0.16 = 2.93$, and relative to $\|\mathbf{X}\|_F^2 = 17.58$ that is 0.167. The two numbers sum to one, which is Corollary 13.3: the proportion of variance explained and the relative approximation error are the same statement

read two ways. ■

Exercise 13.2. Two analysts run `prcomp` on the same data and report first loading vectors $(0.54, 0.58, 0.28, 0.54)$ and $(-0.54, -0.58, -0.28, -0.54)$. Who is right, and what can be said about the component?

Solution. Both. By Proposition 13.4(i) the loading vector is determined only up to sign, and every quantity in the theory is unchanged: the variance $\phi' \mathbf{S} \phi$ is quadratic, the scores change sign with the loadings, and the rank- M reconstruction $\sum_m d_m \mathbf{u}_m \mathbf{v}_m'$ is a product of two things that flip together.

What can be said is invariant under the flip: all four variables load with the same sign and with comparable magnitude, so the first component is an overall level, and any observation scoring high on it is high on all four variables together. What cannot be said is that the component increases with crime, since the other analyst's version decreases with it. ■

Exercise 13.3. Show that the within-cluster sum of squares can be written without reference to a cluster centre, and use the identity to explain why K -means finds spherical clusters.

Solution. Proposition 13.9(i): $\sum_{i \in C} |\mathbf{x}_i - \bar{\mathbf{x}}_C|^2 = (2|C|)^{-1} \sum_{i, i' \in C} |\mathbf{x}_i - \mathbf{x}_{i'}|^2$, proved by expanding both sides and recognising $\sum_i |\mathbf{x}_i|^2 - |C||\bar{\mathbf{x}}_C|^2$.

The criterion is therefore an average squared Euclidean distance within a cluster, and it is small exactly when the cluster is small in every direction at once. An elongated group of points has a large average pairwise squared distance however tightly it hugs its own axis, so K -means prefers to cut it in two, and it prefers clusters of comparable spread because a large loose one contributes heavily. Nothing in Definition 13.8 refers to density or shape. ■

Exercise 13.4. What does Theorem 13.11 promise, what does it not promise, and what follows for practice?

Solution. It promises that each round leaves the criterion no larger, and that the algorithm reaches an assignment it does not change after finitely many rounds. The proof is the two halves of Proposition 13.9(iii) applied in turn, plus the finiteness of the set of partitions.

It does not promise a global minimum. The number of partitions of n points into K non-empty groups is astronomically large, and Lloyd's algorithm visits a decreasing sequence of them from wherever it started. §4.3 runs it from two hundred random starts and reaches several distinct

values.

In practice: use many starts and keep the best, and treat the reported clustering as the best of the starts rather than the solution. ■

Exercise 13.5. Two clusters $A = \{a\}$ and $B = \{b\}$ with $d(a, b) = 2$ merge, and a third cluster $C = \{c\}$ has $d(a, c) = 5$ and $d(b, c) = 9$. Give the next merge height under single, complete and average linkage, and check the reducibility inequality in each case.

Solution. Single: $D(A \cup B, C) = \min\{5, 9\} = 5$. Complete: $\max\{5, 9\} = 9$. Average: $(5 + 9)/2 = 7$.

Reducibility (9) asks for $D(A \cup B, C) \geq \min\{5, 9\} = 5$, and all three give 5, 9 and 7, so all three satisfy it, with single linkage at equality. Each is also at least the height 2 of the merge just made, so the dendrogram does not invert, which is Theorem 13.15.

Centroid linkage would place the merged centroid midway between a and b and its distance to c can be smaller than both 5 and 9 when the geometry allows, which is why it is not covered. ■

Exercise 13.6. Why does single linkage produce long thin clusters, and what does the same property do when one stray observation lies between two groups?

Solution. By Proposition 13.16 two observations share a cluster at height h precisely when a chain of steps each shorter than h connects them. Distance along the chain is never accumulated, so a sequence of closely spaced points joins two ends that are arbitrarily far apart, and the resulting cluster follows whatever path the data lay down.

The same mechanism is the failure. One observation midway between two well-separated groups gives a two-step chain, so the two groups merge at a height set by the stray point alone. Complete linkage is governed by the farthest pair and refuses that merge until the two groups are close throughout, which is why the two linkages disagree most in exactly the cases where it matters. ■

Exercise 13.7. An analyst reports that within-cluster sum of squares falls from 196 at $K = 1$ to 78 at $K = 3$ and 43 at $K = 6$, and concludes that $K = 6$ is best. What is wrong?

Solution. By Proposition 13.12 the optimal within-cluster sum of squares is non-increasing in K and reaches 0 at $K = n$, so the sequence falling is a theorem and not evidence. The conclusion

would be $K = 50$.

The falls themselves can be read: $196 \rightarrow 78$ removes 60% of the total, and $78 \rightarrow 43$ removes another 18%, so the first two splits do most of the work and the picture flattens. That is the elbow, and it is a judgement, not a criterion. It has the same status as the scree plot of §1.3, and for the same reason: with no response there is nothing to hold out. ■

Exercise 13.8. On **USArrests**, the unstandardised first component loads almost entirely on **Assault**. Explain, and say when not standardising would be right.

Solution. The four columns are in different units. §4.2 prints the column standard deviations, and **Assault** at 83 dwarfs **Murder** at 4.4. Principal components maximise variance in the units supplied (Unit 5), so the largest-variance column takes the first direction and carries 97% of the total.

Not standardising is right when the columns are already in the same units and their relative spread is itself the information: repeated measurements of one quantity at different times, or the same price series for several assets. Then rescaling each to unit variance would throw away the fact that one of them moves more than the others. ■

Exercise 13.9. In R: (a) obtain the principal components three ways and reconcile the signs; (b) verify that the truncated singular value decomposition minimises the rank- M approximation error; (c) verify that Lloyd's algorithm never increases the within-cluster sum of squares.

Solution. (a) is §4.1, where the three agree to 10^{-15} in absolute value and `prcomp` and `eigen` return opposite signs on the first component; (b) is the same section, where alternating least squares from twenty random starts never beats the truncation; (c) is §4.3.

The point to watch in (b) is that agreement of the truncation's error with $\sum_{m>M} d_m^2$ only checks the arithmetic of Theorem 13.2; it is the search over other rank- M matrices that checks the optimality, which is the part of the theorem that does work. ■